

B.1 Shur Complement & GP Predictive Distribution

$$\begin{bmatrix} y \\ y_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K(X, X) + \alpha I & K(X, X_*) \\ K(X_*, X) & K(X_*, X_*) \end{bmatrix} \right) \quad \text{where:}$$

y - observed, noisy market targets

$\therefore$ joint covariance matrix

$$\Sigma = \begin{bmatrix} K_{yy} & K_{y*} \\ K_{*y} & K_{**} \end{bmatrix}$$

y_* - unobserved, noise-free predictive targets

$$\text{let } K_{yy} = K(X, X) + \alpha I$$

$$K_{y*} = K(X, X_*)$$

$$K_{*y} = K(X_*, X)$$

$$K_{**} = K(X_*, X_*)$$

We need $P(y_* | y)$

$$M = \begin{bmatrix} I & 0 \\ -K_{*y}K_{yy}^{-1} & I \end{bmatrix} \quad \text{transform matrix}$$

$$\therefore M \begin{bmatrix} y \\ y_* \end{bmatrix} = \begin{bmatrix} I & 0 \\ -K_{*y}K_{yy}^{-1} & I \end{bmatrix} \begin{bmatrix} y \\ y_* \end{bmatrix} = \begin{bmatrix} y \\ y_* - K_{*y}K_{yy}^{-1}y \end{bmatrix}$$

Using the $\text{Cov}(MZ) = M \Sigma M^T$ property, we can show that the data became uncorrelated.

$$\begin{aligned} M \Sigma M^T &= \begin{bmatrix} I & 0 \\ -K_{*y}K_{yy}^{-1} & I \end{bmatrix} \begin{bmatrix} K_{yy} & K_{y*} \\ K_{*y} & K_{**} \end{bmatrix} \begin{bmatrix} I - K_{yy}^{-1}K_{y*} \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} K_{yy} & K_{y*} \\ -K_{*y}K_{yy}^{-1}K_{yy} + K_{y*} & -K_{*y}K_{yy}^{-1}K_{y*} + K_{**} \end{bmatrix} M^T \\ &= \begin{bmatrix} K_{yy} & K_{y*} \\ 0 & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix} \begin{bmatrix} I & -K_{yy}^{-1}K_{y*} \\ 0 & I \end{bmatrix} \quad \text{Shur Complement} \\ &= \begin{bmatrix} K_{yy} & -K_{yy}K_{yy}^{-1}K_{y*} + K_{y*} \\ 0 & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix} = \begin{bmatrix} K_{yy} & 0 \\ 0 & K_{**} - K_{*y}K_{yy}^{-1}K_{y*} \end{bmatrix} \end{aligned}$$

$\therefore y_* - K_{*y}K_{yy}^{-1}y \mid y \sim \mathcal{N}(0, S)$ as $y_* - K_{*y}K_{yy}^{-1}y$ carries statistical independence from y .

Proof: $\text{Cov}(y, y_* - K_{*y}K_{yy}^{-1}y) =$

$$= \text{Cov}(y, y_*) + \text{Cov}(y, -K_{*y}K_{yy}^{-1}y) =$$

$$f_* = K_{xy} K_{yy}^{-1} y \mid y \sim N(0, S)$$

$$f_* \mid y \sim N(K_{xy} K_{yy}^{-1} y, S)$$

$$\therefore \bar{f}_* = K_{xy} K_{yy}^{-1} y \Rightarrow K(X_*, X) [K(X, X) + \alpha I]^{-1} y$$

$$\begin{aligned} \Sigma' = S &\Rightarrow K_{**} - K_{xy} K_{yy}^{-1} K_{yx} = \\ &= K(X_*, X_*) - K(X_*, X) [K(X, X) + \alpha I]^{-1} K(X, X_*) \end{aligned}$$

Q.E.D.

B.2 Matrix calculus of the LM

$$p(y|X, \theta) = (2\pi)^{-\frac{n}{2}} |K_y|^{-\frac{1}{2}} \exp\left(-\frac{1}{2} y^T K_y^{-1} y\right) \quad K_y = K_\theta + \sigma_u^2 I$$

$$\log(p(y|X, \theta)) = \log\left((2\pi)^{-\frac{n}{2}} |K_y|^{-\frac{1}{2}} \exp\left(-\frac{1}{2} y^T K_y^{-1} y\right)\right) \quad \text{Covariance matrix}$$

$$\mathcal{L}(\theta) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log(|K_y|) - \frac{1}{2} y^T K_y^{-1} y$$

$$\frac{\partial}{\partial \theta_j} \mathcal{L}(\theta) = \frac{\partial}{\partial \theta_j} \left(-\frac{n}{2} \log(2\pi) - \frac{1}{2} \log(|K_y|) - \frac{1}{2} y^T K_y^{-1} y \right)$$

$$= -\frac{\partial}{\partial \theta_j} \left(\frac{1}{2} \log(|K_y|) \right) - \frac{1}{2} y^T \frac{\partial}{\partial \theta_j} (K_y^{-1}) y$$

$$= -\frac{1}{2} \cdot \frac{1}{|K_y|} \cdot \frac{\partial}{\partial \theta_j} |K_y| - \frac{1}{2} y^T \left(-K_y^{-1} \frac{\partial}{\partial \theta_j} K_y^{-1} K_y^{-1} \right) y$$

$$= -\frac{1}{2} \cdot \frac{1}{|K_y|} \cdot (|K_y| \operatorname{tr}(K_y^{-1} \frac{\partial}{\partial \theta_j} K_y)) + \frac{1}{2} y^T \dots$$

$$\frac{\partial}{\partial \theta_j} \mathcal{L}(\theta) = -\frac{1}{2} \operatorname{tr}(K_y^{-1} \frac{\partial}{\partial \theta_j} K_y) + \frac{1}{2} y^T K_y^{-1} \frac{\partial}{\partial \theta_j} K_y K_y^{-1} y$$

$$= \frac{1}{2} \alpha^T \frac{\partial}{\partial \theta_j} K_y \alpha - \frac{1}{2} \operatorname{tr}(K_y^{-1} \frac{\partial}{\partial \theta_j} K_y)$$

$$= \frac{1}{2} \operatorname{tr}(\alpha \alpha^T \frac{\partial}{\partial \theta_j} K_y) - \frac{1}{2} \operatorname{tr}(K_y^{-1} \frac{\partial}{\partial \theta_j} K_y)$$

$$= \frac{1}{2} \operatorname{tr}((\alpha \alpha^T - K_y^{-1}) \frac{\partial}{\partial \theta_j} K_y)$$

$$\text{Since } \frac{\partial}{\partial \theta_j} (K_y K_y^{-1}) = \frac{\partial K_y}{\partial \theta_j} K_y^{-1} + K_y \frac{\partial K_y^{-1}}{\partial \theta_j} = 0$$

that follows →

$$\frac{\partial K_y^{-1}}{\partial \theta_j} = -K_y^{-1} \frac{\partial K_y}{\partial \theta_j} K_y^{-1}$$

let

$$\alpha = K_y^{-1} y$$

$$\alpha^T = y^T (K_y^{-1})^T$$

$$= y^T K_y^{-1}$$

cuz $(K_y^{-1})^T = K_y^{-1}$ as covariance matrix are symmetric

Q.E.D.

B.3 Analytical Derivative of the Matérn $5/2$ Kernel

$$\bar{f}_x = K_x^T (K_{yy})^{-1} y = K_x^T (K_{yy})^{-1} y \\ = u_x^T \alpha$$

$$u_x^T = K(X, x_x) \\ \alpha = K_{yy}^{-1} y$$

$$\frac{\partial \bar{f}_x}{\partial x_{x,i}} = \left(\frac{\partial u_x}{\partial x_{x,i}} \right)^T \alpha$$

$$\frac{\partial^2 \bar{f}_x}{\partial x_{x,i}^2} = \left(\frac{\partial^2 u_x}{\partial x_{x,i}^2} \right) \alpha$$

$$u_{5/2}(r) = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2 \right)$$

$$\cdot \exp(-\sqrt{5}r)$$

$$\frac{\partial u_{5/2}}{\partial x_{x,i}} = \frac{\partial u_{5/2}}{\partial r} \frac{\partial r}{\partial x_{x,i}}$$

$$r(X, x_x) = \sqrt{\frac{(x_{x,1} - x_{x,1})^2}{c_1^2} + \frac{(x_{x,2} - x_{x,2})^2}{c_2^2}}$$

$$x_1 = K(\text{strike}), x_2 = T(\text{maturity})$$

$$\frac{\partial u_{5/2}}{\partial r} = \frac{\partial}{\partial r} \left(\sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2 \right) e^{-\sqrt{5}r} \right)$$

$$= \frac{\partial}{\partial r} \left(\sigma_f^2 e^{-\sqrt{5}r} + \sqrt{5}r \sigma_f^2 e^{-\sqrt{5}r} + \frac{5}{3}r^2 \sigma_f^2 e^{-\sqrt{5}r} \right)$$

$$= -\sqrt{5} \sigma_f^2 e^{-\sqrt{5}r} + \sqrt{5} \sigma_f^2 \left(e^{-\sqrt{5}r} + (-\sqrt{5}r) e^{-\sqrt{5}r} \right) \\ + \frac{5}{3} \sigma_f^2 \left(2r e^{-\sqrt{5}r} + (-\sqrt{5}r^2) e^{-\sqrt{5}r} \right)$$

$$= -\sqrt{5} \sigma_f^2 e^{-\sqrt{5}r} + \sqrt{5} \sigma_f^2 e^{-\sqrt{5}r} - 5 \sigma_f^2 r e^{-\sqrt{5}r}$$

$$+ \frac{10}{3} \sigma_f^2 r e^{-\sqrt{5}r} - \frac{5\sqrt{5}}{3} \sigma_f^2 r^2 e^{-\sqrt{5}r}$$

$$= \sigma_f^2 e^{-\sqrt{5}r} \left(\frac{10}{3}r - 5r - \frac{5\sqrt{5}}{3}r^2 \right)$$

$$-\frac{5}{3}r - \frac{5\sqrt{5}}{3}r^2$$

$$= \sigma_f^2 e^{-\sqrt{5}r} \cdot -\frac{5}{3} (1 + \sqrt{5}r^2)$$

$$\frac{\partial r}{\partial x_{x,i}} = \frac{\partial}{\partial x_{x,i}} \left(\sqrt{\frac{(x_1 - x_{x,1})^2}{c_1^2} + \frac{(x_2 - x_{x,2})^2}{c_2^2}} \right)$$

$$\frac{\partial}{\partial x_{k,i}} = \frac{\partial}{\partial x_{k,i}} \left(\frac{(x_{k,1} - x_1)^2}{e_1^2} + \frac{(x_{k,2} - x_2)^2}{e_2^2} \right)^{1/2} \quad \sqrt{x} = \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2r}$$

$$\frac{\partial r}{\partial x_{k,i}} = \frac{(x_{k,1} - x_1)^2}{e_1^2} + \frac{(x_{k,2} - x_2)^2}{e_2^2} = \frac{2(x_{k,1} - x_1) \cdot 1}{e_1^2}$$

$$= \frac{1}{2r} \left(\frac{2(x_{k,1} - x_1)}{e_1^2} \right) = \frac{x_{k,1} - x_1}{re_1^2}$$

$$\frac{\partial u_{S_2}}{\partial x_{k,i}} = -\frac{5}{3} \sigma_f^2 (1 + \sqrt{5}) e^{-\sqrt{5}r} \cdot \frac{x_{k,i} - x_i}{re_1^2}$$

$$\frac{\partial^2 u_{S_2}}{\partial x_{k,i}^2} = -\frac{5}{3} \sigma_f^2 (1 + \sqrt{5}) e^{-\sqrt{5}r} \cdot D_k \quad \text{let } \frac{x_{k,i} - x_i}{re_1^2} = D_k$$

$$= -\frac{5}{3} \sigma_f^2 \left[\frac{\partial}{\partial x_{k,i}} \left((1 + \sqrt{5}r) e^{-\sqrt{5}r} \right) \cdot D_k + (1 + \sqrt{5}r) e^{-\sqrt{5}r} \cdot \frac{\partial D_k}{\partial x_{k,i}} \right]$$

$$\text{let } u(r) = (1 + \sqrt{5}r) e^{-\sqrt{5}r}, \quad \frac{\partial u}{\partial r} = \sqrt{5} e^{-\sqrt{5}r} - \sqrt{5} (1 + \sqrt{5}r) e^{-\sqrt{5}r} \cdot \frac{\partial r}{\partial x_{k,i} \sqrt{5}}$$

$$= -5 e^{-\sqrt{5}r} \cdot D_k$$

$$\text{and } \frac{\partial D_k}{\partial x_{k,i}} = \frac{1}{e_i^2}$$

$$\therefore \frac{\partial^2 u_{S_2}}{\partial x_{k,i}^2} = -\frac{5}{3} \sigma_f^2 \left(-5 e^{-\sqrt{5}r} D_k^2 + (1 + \sqrt{5}r) e^{-\sqrt{5}r} \cdot \frac{1}{e_i^2} \right)$$

Q.E.D.

B.4 Breeden - Litzenberger Density and Static Arb.

$$C(K,T) = e^{-rT} \mathbb{E}^Q(\max(S_T - K, 0)) = e^{-rT} \int_0^{\infty} \max(S_T - K, 0) \cdot p(S_T) dS_T$$

max() function - $\max(S_T - K, 0)$

- if $S_T < K$, then $S_T - K$ is negative and $\max(S_T - K, 0) = 0$
- if $S_T \geq K$, then $S_T - K$ is positive and $\max(S_T - K, 0) = S_T - K$

$$C(K,T) = e^{-rT} \left(\int_0^K 0 \cdot p(S_T) dS_T + \int_K^{\infty} (S_T - K) p(S_T) dS_T \right)$$

$$= e^{-rT} \left(0 + \int_K^{\infty} (S_T - K) \cdot p(S_T) dS_T \right)$$

$\therefore$

$$C(K,T) = e^{-rT} \int_K^{\infty} (S_T - K) \cdot p(S_T) dS_T \quad \leftarrow \text{standard risk-neutral pricing f.}$$

$$\frac{d}{dK} C(K,T) = \frac{d}{dK} \left(e^{-rT} \int_K^{\infty} (S_T - K) \cdot p(S_T) dS_T \right)$$

$\frac{d}{dK} 0 = 0, \frac{d}{dK} K = 1$
 $S_T = K \rightarrow (K - K) p(K) \cdot 1 = 0$

$$= e^{-rT} \int_K^{\infty} \frac{\partial}{\partial K} ((S_T - K) \cdot p(S_T)) dS_T$$

$$= e^{-rT} \int_K^{\infty} (-1) \cdot p(S_T) dS_T \quad \leftarrow \text{discounted survival func.}$$

$$\frac{d^2}{dK^2} C(K,T) = -e^{-rT} \frac{\partial}{\partial K} \int_K^{\infty} p(S_T) dS_T = -e^{-rT} (-p(K))$$

$$= e^{-rT} p(K)$$

$\therefore p(K) = e^{-rT} \frac{\partial^2 C}{\partial K^2}$ since $p(K) \geq 0$ and $e^{-rT} \geq 0$

$$\frac{\partial^2 C}{\partial K^2} \geq 0$$

Q.E.D.

B.5 Dupire's Equation from the Fokker-Planck - Kolmogorov

$$C(K, T) = e^{-rT} \int_K^{\infty} (S - K) \cdot p(S, T) dS$$

$$\begin{aligned} \frac{\partial}{\partial T} C(K, T) &= \frac{\partial}{\partial T} \left(e^{-rT} \int_K^{\infty} (S - K) p(S, T) dS \right) \\ &= -r e^{-rT} \int_K^{\infty} (S - K) p(S, T) dS + e^{-rT} \int_K^{\infty} (S - K) \frac{\partial p}{\partial T} dS \\ &= -rC + e^{-rT} \int_K^{\infty} (S - K) \frac{\partial p}{\partial T} dS \quad \leftarrow \text{Insert } \frac{\partial p}{\partial T}, \text{ the FPK eq.} \\ &= -rC + \left[e^{-rT} \int_K^{\infty} (S - K) \frac{1}{2} \frac{\partial^2}{\partial S^2} (\sigma_L^2 S^2 p) dS \right] - \left[e^{-rT} \int_K^{\infty} (S - K) \frac{\partial}{\partial S} (rS p) dS \right] \end{aligned}$$

I. $u = (S - K) \rightarrow du = dS$

$$dV = \frac{\partial^2}{\partial S^2} (\sigma_L^2 S^2 p) dS \rightarrow v = \frac{\partial}{\partial S} (\sigma_L^2 S^2 p)$$

$$\begin{aligned} \int_K^{\infty} (S - K) \frac{1}{2} \frac{\partial^2}{\partial S^2} (\sigma_L^2 S^2 p) dS &= (S - K) \frac{\partial}{\partial S} (\sigma_L^2 S^2 p) \Big|_K^{\infty} - \int_K^{\infty} \frac{\partial}{\partial S} (\sigma_L^2 S^2 p) dS \\ &= 0 - \int_K^{\infty} \frac{\partial}{\partial S} (\sigma_L^2 S^2 p) dS = -\sigma_L^2 S^2 p \Big|_K^{\infty} \end{aligned}$$

$$= - (0 - \sigma_L^2(K, T) K^2 p(K, T)) = \sigma_L^2(K, T) K^2 p(K, T)$$

II. $u = (S - K) \rightarrow du = dS$

$$dV = \frac{\partial}{\partial S} (rS p) dS \rightarrow v = rS p \quad \nearrow \text{as } S \rightarrow \infty, p(S, T) \rightarrow 0$$

$$\begin{aligned} \int_K^{\infty} (S - K) \frac{\partial}{\partial S} (rS p) dS &= (S - K) rS p \Big|_K^{\infty} - \int_K^{\infty} rS p dS \\ &= 0 - \int_K^{\infty} rS p dS = - \int_K^{\infty} rS p dS \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial T} C(K, T) &= -rC + e^{-rT} \left(\frac{1}{2} \sigma_L^2(K, T) K^2 p(K, T) \right) - e^{-rT} \left(- \int_K^{\infty} rS p dS \right) \\ &= -rC + \frac{1}{2} \sigma_L^2(K, T) \cdot K^2 \cdot (e^{-rT} p(K, T)) + r e^{-rT} \int_K^{\infty} S p dS \end{aligned}$$

from B.4, $\bar{e}^{rT} p(K, T) = \frac{\partial^2 \mathcal{C}}{\partial K^2}$ and $\frac{\partial \mathcal{C}}{\partial u} = -\bar{e}^{rT} \int_u^\infty p ds$

$$\begin{aligned} \frac{\partial}{\partial T} C(K,T) &= -rC + \frac{1}{2}\sigma_L^2(K,T)K^2 \frac{\partial^2 C}{\partial K^2} + re^{-rT} \int_K^\infty S p dS \\ &= " " + re^{-rT} \int_K^\infty (S-K+K)p dS \\ &= " " + re^{-rT} \left(\int_K^\infty (S-K)p dS + K \int_K^\infty p dS \right) \\ &= -rC + \frac{1}{2}\sigma_L^2(K,T)K^2 \frac{\partial^2 C}{\partial K^2} + r(C - K \frac{\partial C}{\partial K}) \end{aligned}$$

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma_L^2(K, T) L^2 \frac{\partial^2 C}{\partial K^2} - r K \frac{\partial C}{\partial K}$$

$$\frac{\partial C}{\partial T} - rK \frac{\partial C}{\partial K} = \frac{1}{2} \sigma_L^2(K, T) K^2 \frac{\partial^2 C}{\partial K^2}$$

$$\frac{\frac{\partial C}{\partial T} - rK \frac{\partial C}{\partial K}}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}} = \sigma_L^2(K, T)$$

D. E. D.

B.6 Mercer's Theorem and the RKHS

Let X be a compact space and $k \in C(X \times X)$ be symmetric, PSD kernel.

$T_k: L^2(X) \rightarrow L^2(X)$ such that:

$$(T_k f)(x) = \int_X k(x, x') f(x') dx'$$

By Mercer's Theorem, the eigenvalue problem:

$\int_X k(x, x') \phi_i(x') dx' = \lambda_i \phi_i(x)$ results in an orthonormal basis of eigenfunctions $\phi_i|_{i=1}^{\infty}$ and eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$.

Additionally, $k(x, x') = \sum_{i=1}^{\infty} \lambda_i \phi_i(x) \phi_i(x')$, and we define $\mathcal{H}_k = \text{span } \phi_i|_{i=1}^{\infty}$. So, we pick two of these eigenfunctions:

$$f(x) = \sum_{i=1}^{\infty} c_i \phi_i(x) \quad \text{and} \quad g(x) = \sum_{i=1}^{\infty} d_i \phi_i(x)$$

To prove the reproducing property, we evaluate the dot product $\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k}$:

$$\begin{aligned} \langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} &= \sum_{i=1}^{\infty} c_i \underbrace{(\lambda_i \phi_i(x))}_{\lambda_i} \\ &= \sum_{i=1}^{\infty} c_i \phi_i(x) \end{aligned}$$

$$\langle f, g \rangle_{\mathcal{H}_k} = \sum_{i=1}^{\infty} \frac{c_i d_i}{\lambda_i}$$

$$\langle f, k(\cdot, x) \rangle_{\mathcal{H}_k} = f(x)$$



So we proved that $\mathcal{H}_k$ is a valid Hilbert space.

The RKHS norm $\|f\|_{\mathcal{H}_K}$ is computed using the identity $\langle f, f \rangle_{\mathcal{H}_K} = \|f\|_{\mathcal{H}_K}^2$:

$$\|f\|_{\mathcal{H}_K}^2 = \sum_{i=1}^{\infty} \frac{c_i c_i}{\lambda_i} = \sum_{i=1}^{\infty} \frac{c_i^2}{\lambda_i}$$

The objective functional $J(f)$ is

$$\begin{aligned} J(f) &= \sum_{j=1}^N (y_j - f(x_j))^2 + \sigma_n^2 \|f\|_{\mathcal{H}_K}^2 \\ &= \sum_{j=1}^N \left(y_j - \sum_{i=1}^{\infty} c_i \phi_i(x_j) \right)^2 + \sigma_n^2 \sum_{i=1}^{\infty} \frac{c_i^2}{\lambda_i} \end{aligned}$$

Q.E.D.

B.7 Fisher Information Matrix

$$l(\theta) = -\frac{1}{2} y^T K^{-1} y - \frac{1}{2} \log |K| - \frac{n}{2} \log(\kappa \sigma^2)$$

$$\frac{\partial l}{\partial \sigma_i^2} = \frac{1}{2} y^T K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1} y - \frac{1}{2} \text{tr}(K^{-1} \frac{\partial K}{\partial \sigma_i^2}) \quad \text{from B.2}$$

$$\frac{\partial^2 l}{\partial \sigma_i^2 \partial \sigma_j^2} = \frac{\partial}{\partial \sigma_i^2} \left(\frac{1}{2} y^T K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1} y - \frac{1}{2} \text{tr}(K^{-1} \frac{\partial K}{\partial \sigma_i^2}) \right)$$

$$= \frac{1}{2} y^T \left(\frac{\partial}{\partial \sigma_j^2} \left(K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1} \right) \right) y - \frac{1}{2} \text{tr} \left(\frac{\partial}{\partial \sigma_j^2} \left(K^{-1} \frac{\partial K}{\partial \sigma_i^2} \right) \right)$$

$$= \frac{1}{2} y^T \left(\left(-K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \right) \frac{\partial K}{\partial \sigma_i^2} K^{-1} + K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} K^{-1} + \right.$$

$$\left. + K^{-1} \frac{\partial K}{\partial \sigma_i^2} \left(-K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \right) \right) y - \frac{1}{2} \text{tr} \left(\left(-K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \right) \frac{\partial K}{\partial \sigma_i^2} + \right.$$

cuz they're transposes of each other
 term ① = ③, $\therefore -2K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1}$

$$= \frac{1}{2} y^T \left(-K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1} + K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} K^{-1} - K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \right) y$$

$$+ \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} - K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} \right)$$

Fisher Info Matrix $I_{ij} = -\mathbb{E}_y \left(\frac{\partial^2 l}{\partial \sigma_i^2 \partial \sigma_j^2} \right)$

$$= -\mathbb{E}_y \left(\frac{1}{2} y^T \left(-2K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} K^{-1} + K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} K^{-1} \right) y + \right.$$

$$\left. + \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} \right) - \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} \right) \right) \quad \mathbb{E}[y^T A y] = \text{tr}(AK)$$

$$= -\frac{1}{2} \text{tr} \left(\left(-2K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} + K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} \right) I \right)$$

$$\left[-\frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} \right) + \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} \right) \right] \leftarrow \text{deterministic}$$

$$= \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} \right) - \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} \right)$$

$$- \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j^2} K^{-1} \frac{\partial K}{\partial \sigma_i^2} \right) + \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial^2 K}{\partial \sigma_i^2 \partial \sigma_j^2} \right)$$

→

$$\begin{aligned}
 I_{ij} &= \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j} K^{-1} \frac{\partial K}{\partial \sigma_i} \right) - \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j} K^{-1} \frac{\partial K}{\partial \sigma_i} \right) \\
 &= \frac{1}{2} \text{tr} \left(K^{-1} \frac{\partial K}{\partial \sigma_j} K^{-1} \frac{\partial K}{\partial \sigma_i} \right)
 \end{aligned}$$

Q.E.D.